EXP. 2
23/7/24

Study of Different Types of Network Cables

AIM:

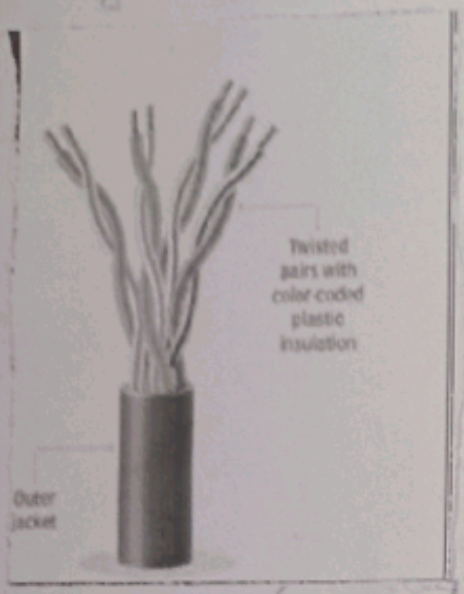
To study and understand different type of network cable, To make your own Ethernet cross over cable/ straight cable.

a) Understand different types of network cable

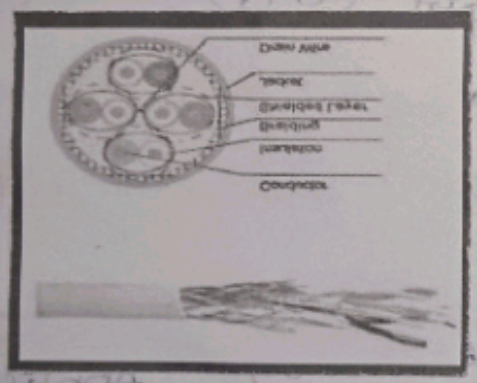
Different type of cables used in networking :

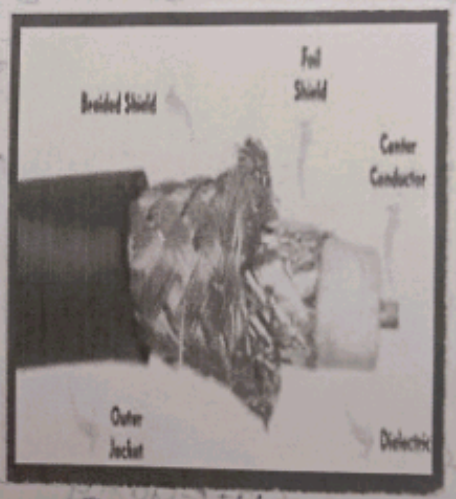
i) Unshielded Twisted pair (UTP) Cable
ii) Shielded Twisted pair (STP) Cable
iii) Coaxial cable
iv) Fibre optic cable

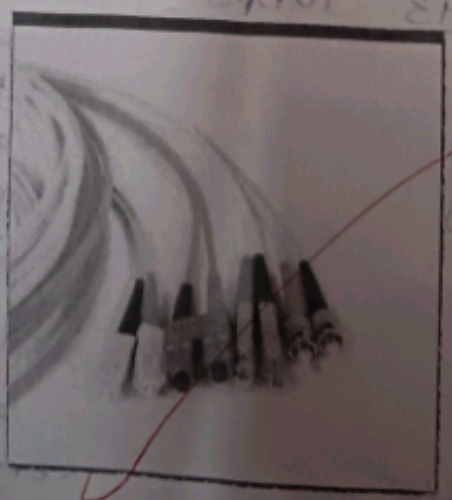
Cable Type	Category	Maximum Data Transmissio	Advantages/ Disadvantages	Application
UTP	Category 3	10bps	Advantages: * Cheaper in cost * Easy to Install as they have a Smaller overall diameter. Disadvantages: * More prone to (EMI) Electromagnatic Interference & noise	10 Base-T Ethernet
	Category 5	upto 100 Mbps		Fast Ethernet, Gigabit Ethernet
	category 5E	1 Gbps		


Twisted pairs with color-coded plastic insulation
Outer jacket

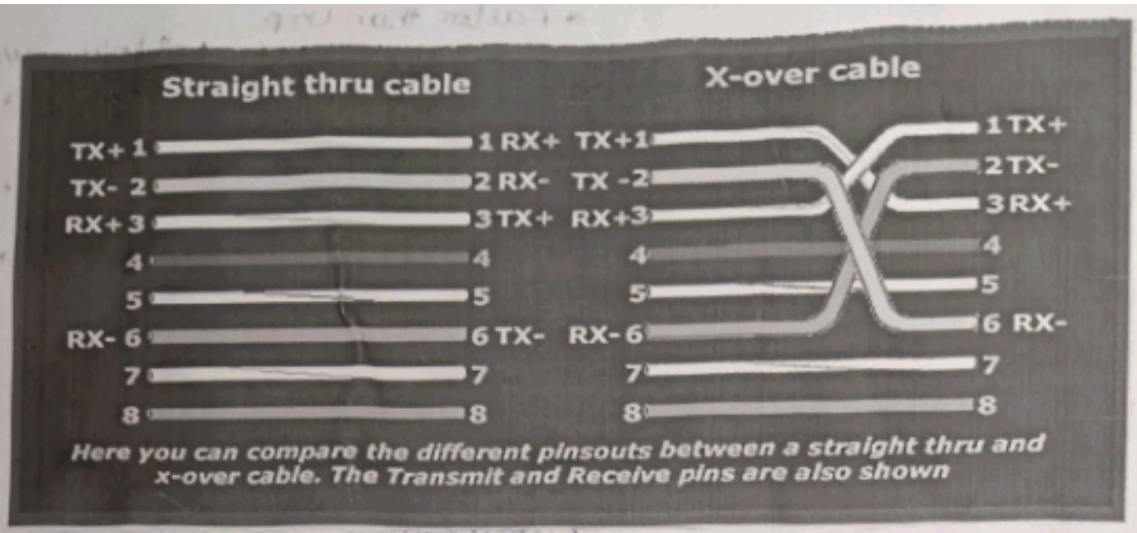

Twisted pairs with color-coded plastic insulation
Pair shields
Overall shield
Outer jacket
Shielded twisted pair




Braided Shield
Foil Shield
Center Conductor
Outer Jacket
Dielectric

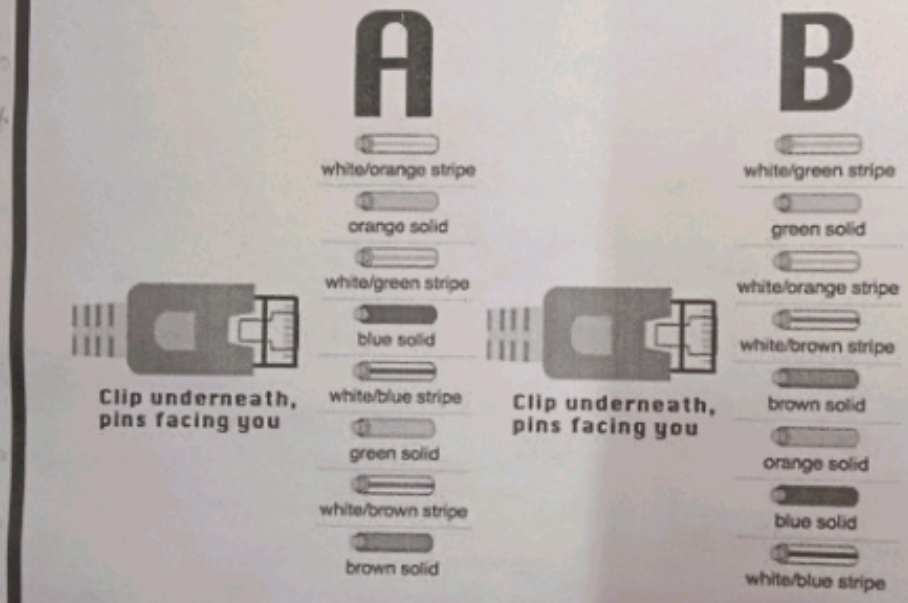


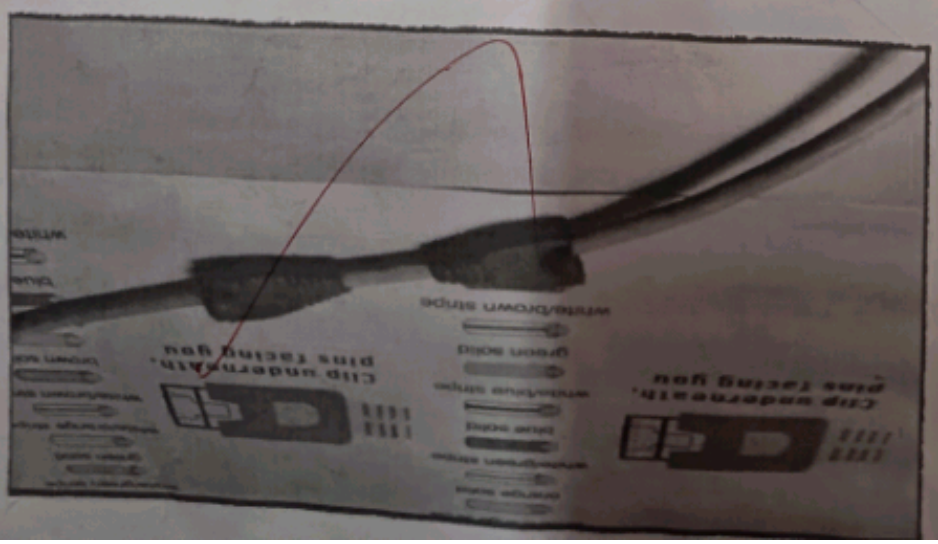
STP	Category 6, 6a	10 Gbps	Advantages: * Shielded * Faster than UTP * Less susceptible to noise and interference Disadvantages: * Expensive * Greater installation effort	Gigabit Ethernet, 10G Ethernet (55m) widely used in data centres
SSTP	category 7	10Gbps		Gigabit Ethernet, 10G Ethernet (100m)
Coaxial Cable	RG-6 RG-6a RG-11	10 – 100 Mbps	Advantages: * High bandwidth * Immune to interference * low loss bandwidth * versatile Disadvantages: * Limited distance * cost * Size is bulky	Speed of signal is 500m Television network High speed internet connections
Fiber Optics Cable	Single mode Multi mode	100 Gbps	Advantages: * High Speed * High bandwidth * High Security * Long Distance Disadvantages: * Expensive * Requires skilled installer	Maximum distance of fiber optics cable is around 100m.



Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

Straight through network cable: both sides should be A
Crossover cable: One side A, one side B





Make your own Ethernet Cross-over cable/ Straight cable

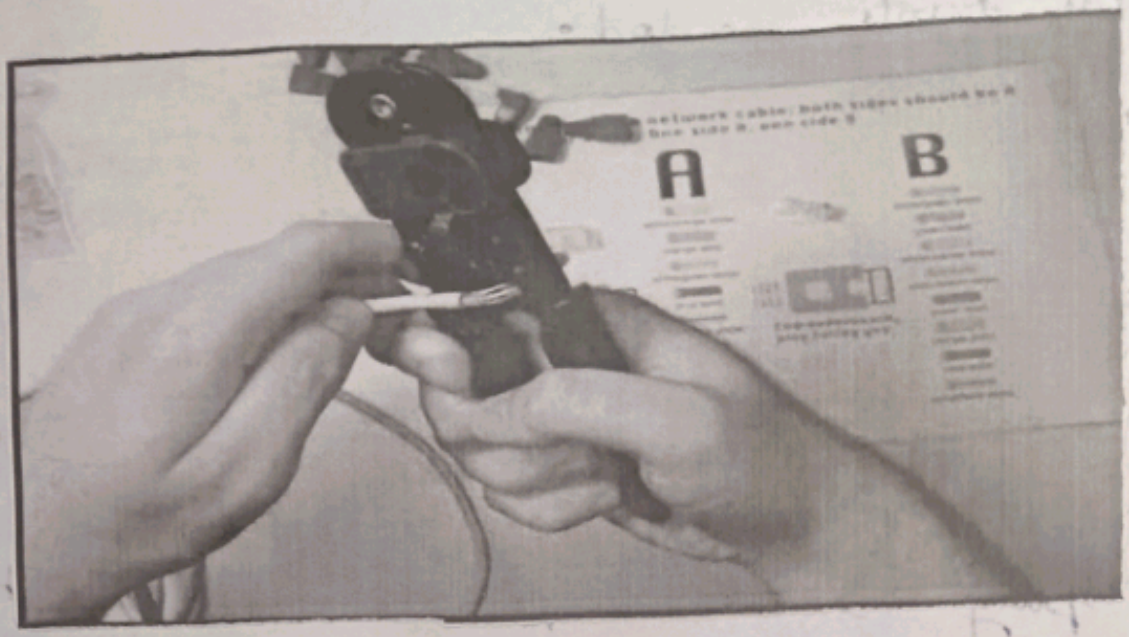
Tools and parts needed:

* Ethernet cabling. CAT5E is certified for gigabit support, but CAT5 cabling working as well, just over shorter distances.
* A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.
* Two RJ45 plugs.
* Optional two plug shields.

Difference between crossover cable and straight cab

Step1:
To start construction of the device, begin by threading shields onto the cable.

Step2:
Next, strip approximately 1.5cm of cable shielding from both ends. The crimping tool has a round area to complete this task.


A
B


A
orange solid
white/green stripe
blue solid

Step 3:

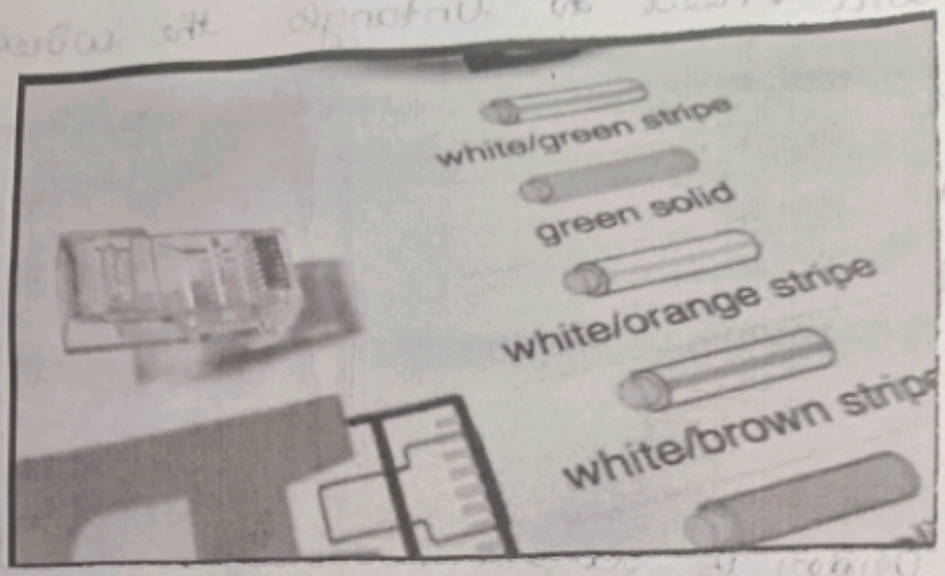
After, you will need to untangle the wires; there should be four "twisted pairs". Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.

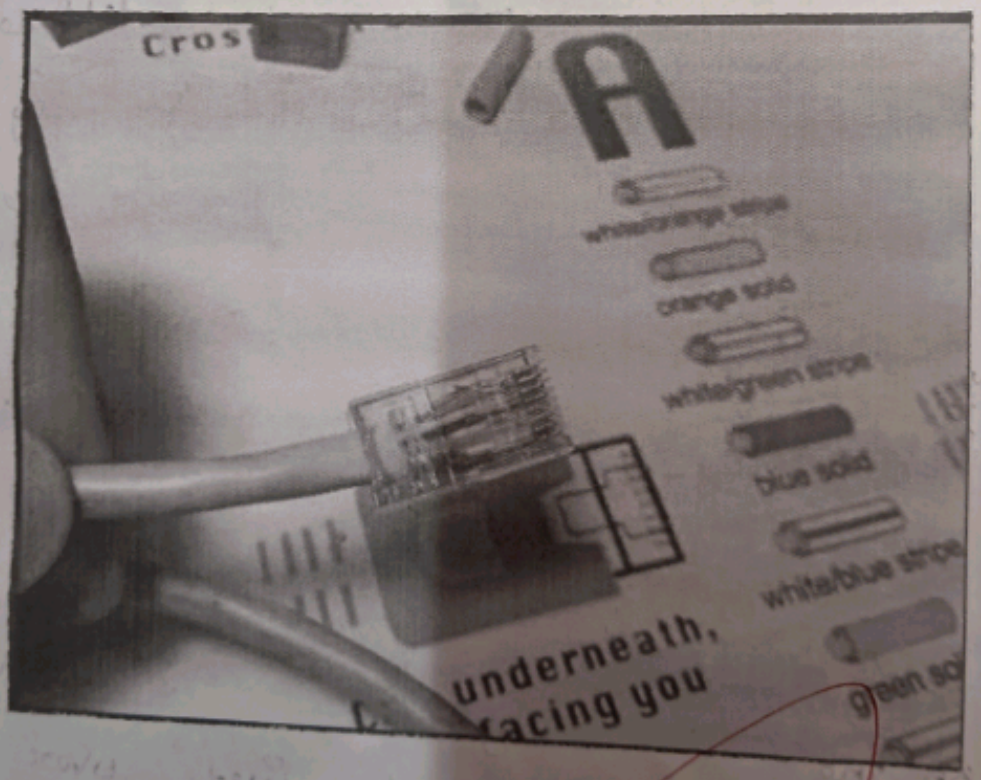
Step 4:

Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others, snip them back to create an even level. The different aspect is placing these into the RJ45 plug without messing up the order. To do so, hold the plug with the clips side facing away from you and have the gold pins facing toward you.

Step 5:

Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding, and if it isn't, that means that you stripped off too too much shielding. Simply snip the cables back a little more.


white/green stripe
green solid
white/orange stripe
white/brown stripe


Cros
A
underneath,
facing you

Step 6:

After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

It should be sharped correctly, but pushing too hard can crack the fragile plastic plug.

Step 7:

Lastly, repeat for the other end using diagram B (to make a crossover cables) using diagram A (to make a straight through cable)

To test it, plug it in and attempt to connect two devices directly.

90

RESULT:

Thus, different types of network cables were studied successfully and straight Through and Cross over Ethernet cables were prepared and tested successfully.